Name of the Student	**P Surendra Kumar Reddy**
Reg. No	**192425224**
GitHub Link	**https://github.com/suri307/MLA0403-DEEP-LEARNING/tree/main/ASSESSMENTS**

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL 3
Comparative Analysis

COURSE NAME: DEEP LEARNING **COURSE CODE:MLA0403**
SLOT :A

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25% **Duration: 90 minutes**

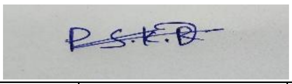
Assessment Tool Weightage: 40% **Marks: 25**

Code of Conduct:

I P Surendra Kumar Reddy (192425224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Name of the student: P Surendra Kumar Reddy**



Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation *(1)*	Total (10)
Weightage	*20 %*	*25%*	*25%*	*20 %*	*10 %*	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty: ***Total Marks :***

QUESTIONS: 05 **Total Marks:25**

Q1. **CNN vs Fully Connected Neural Network**

A medical imaging organization wants to classify X-ray images using deep learning. Compare a CNN with a traditional fully connected neural network in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Solution:

1. Problem Understanding

For X-ray image classification, the model must identify visual patterns such as edges, textures, shapes, and disease-related regions. Two possible approaches are a traditional Fully Connected Neural Network (FCNN) and a Convolutional Neural Network (CNN).A CNN is more suitable because it preserves the spatial arrangement of pixels and automatically learns hierarchical visual features.

2. Architectural Comparison

Parameter	Fully Connected Neural Network	CNN
Basic structure	Every neuron connects to all neurons in next layer	Uses convolution, pooling and FC layers
Input handling	Image is usually flattened	Image remains as 2D/3D structure
Spatial information	Mostly lost after flattening	Preserved
Feature extraction	Requires large number of learned connections	Automatically extracts local features
Parameters	Very high	Comparatively lower
Computation	Expensive for large images	More efficient due to local connections
Image suitability	Poor for large images	Excellent
Translation handling	Weak	Better due to shared filters and pooling

3. Role of CNN Components

Convolutional Layer

The convolutional layer applies small filters to an image to detect local patterns.

For example:

X-ray → Convolution → Edge/Texture Detection → Feature Map

A filter can detect:

- Edges
- Curves

- Textures
- Shapes
- Abnormal regions

Filters

Filters are small matrices such as 3×3 or 5×5. During training, CNN learns the most useful filter values automatically.

Early filters may identify simple edges, while deeper filters can recognize complex structures.

Pooling Layer

Pooling reduces the spatial size of feature maps.

For example:

28 × 28 → Max Pooling → 14 × 14

This:

- Reduces computation
- Reduces memory usage
- Retains important features
- Provides some translation tolerance

4. Parameter Analysis

Suppose an X-ray image is **224 × 224 × 3**.

The number of input values is:

If these are fully connected to only 1,000 neurons:

This requires approximately **150 million weights** for just one connection layer.

A convolution layer with 64 filters of size 3×3 requires:

weights, excluding biases.

Therefore, CNNs can dramatically reduce the number of parameters.

5. Critical Evaluation

A fully connected network treats pixels largely as independent input values and does not naturally understand that neighboring pixels form meaningful structures.

CNNs exploit **local connectivity and parameter sharing**, making them much more appropriate for X-ray images.

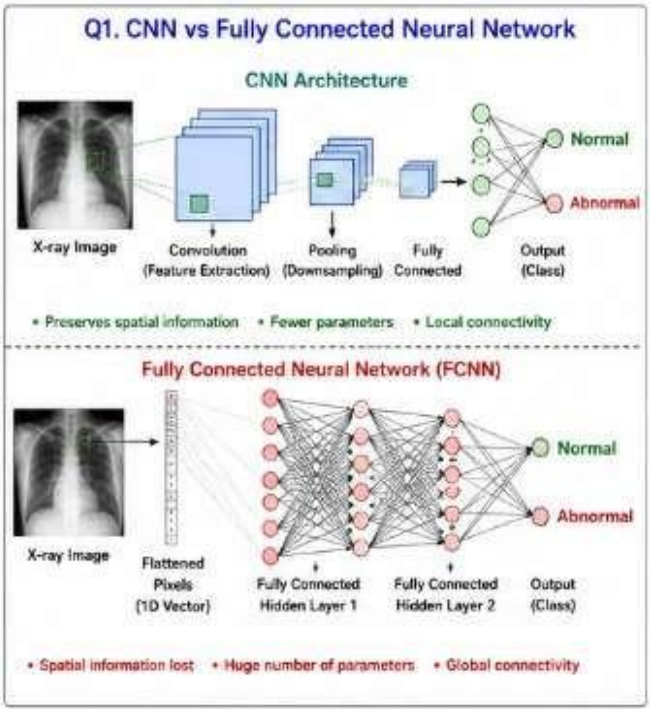


Figure 1: CNN Vs Fully Connected Neural Network

Conclusion

CNN is the preferred architecture for medical image classification because it:

- Preserves spatial information
- Automatically learns image features
- Uses fewer parameters
- Requires less memory
- Extracts hierarchical features
- Provides better performance on image data

Therefore, CNN provides a more efficient and scalable solution for automated X-ray cla

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of convolutional layers, pooling layers, activation layers, and fully connected layers in terms of feature extraction, spatial information, computational requirements, and classification. Further compare early-layer filters and deeper-layer filters based on the type of features they learn, and justify how these layers collectively improve defect detection.

Solution:

1. Problem Understanding

In industrial defect detection, the CNN must identify abnormalities such as:

- Cracks
- Scratches
- Holes
- Surface irregularities
- Shape distortions

Different CNN layers contribute different levels of feature extraction.

2. Comparison of CNN Layers

Layer	Main Function	Spatial Information	Computational Requirement	Role in Defect Detection
Convolution	Extract features	Preserves spatial structure	Moderate	Detects edges, textures and patterns
Pooling	Reduces dimensions	Partially reduces spatial detail	Low	Keeps important features while reducing computation
Activation	Adds non-linearity	Maintains dimensions	Low	Helps model complex patterns
Fully Connected	Performs classification	Spatial arrangement largely removed	High	Produces final class prediction

3. Convolutional Layer

The convolution layer applies filters to the input image.

For example:

Product Image → Filter → Feature Map

A filter can identify:

- Horizontal edges

- Vertical edges
- Corners
- Textures
- Cracks

The convolution operation can be represented as:

where * represents convolution.

4. Pooling Layer

Pooling reduces the size of feature maps.

Example:

using 2×2 max pooling.

Benefits:

- Reduces computation
- Reduces memory
- Removes less important information
- Helps tolerate small shifts in defect location

5. Activation Layer

A common activation function is ReLU:

It introduces non-linearity, allowing the network to learn complicated defect patterns.

6. Fully Connected Layer

The final feature representation is passed to fully connected layers.

For binary defect classification:

Feature Maps → Flatten → Fully Connected → Output

Output could be:

- **0 = Normal**
- **1 = Defective**

7. Early vs Deep Filters

Feature	**Early Filters**	**Deeper Filters**
Feature complexity	Simple	Complex
Detects	Edges, lines	Shapes, textures and object parts
Receptive field	Small	Larger
Level of abstraction	Low	High
Example	Crack edges	Complete crack pattern

Hierarchical learning

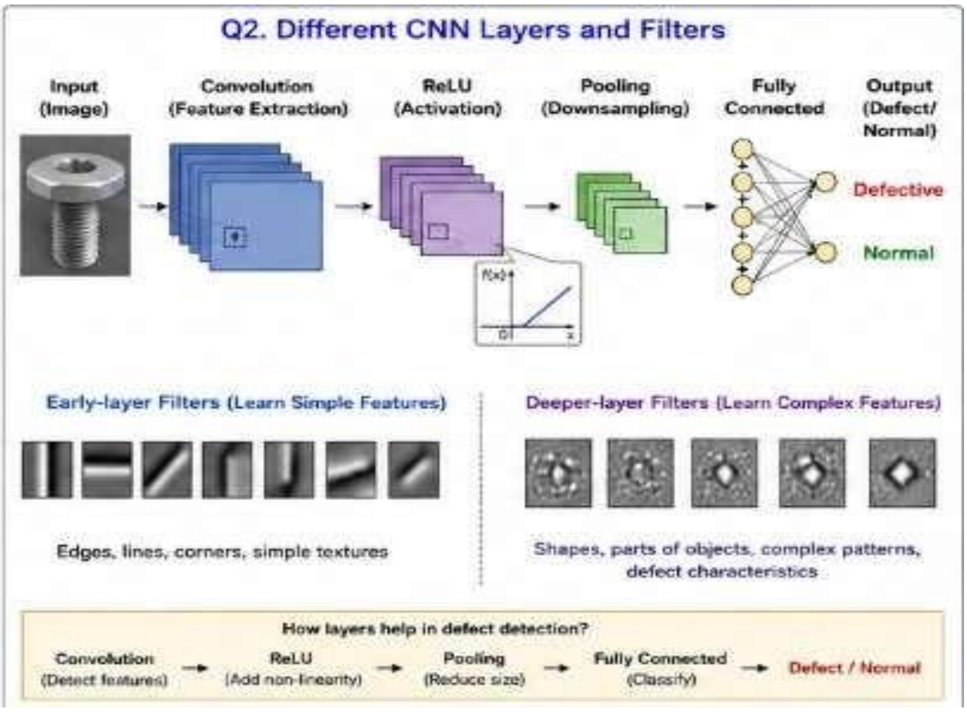


Figure 2: Different CNN Layers and Filters

8. Critical Evaluation

Using only early layers would identify basic edges but might not distinguish a genuine manufacturing defect from normal surface texture.

Deeper layers combine simpler features into meaningful defect patterns.

Conclusion

The layers work collectively:

Convolution → Feature Extraction

ReLU → Non-linearity

Pooling → Dimension Reduction

Fully Connected → Classification

Therefore, combining these layers allows CNNs to detect both simple and complex industrial defects efficiently and accurately.

Q3. **Parameter Sharing vs Independent Weights**

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare parameter sharing in CNNs with independent weights in fully connected networks in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Solution:

1. Problem Understanding

A surveillance system may need to process thousands of high-resolution images. Therefore, the model must have:

- Low memory requirements
- Efficient computation
- Fast training
- Good feature detection

CNN parameter sharing provides a major advantage over fully connected networks.

2. Comparison

Parameter	**CNN with Parameter Sharing**	**Fully Connected Network**
Weights	Same filter reused across image	Different weights for connections
Number of parameters	Low	Very high
Memory requirement	Lower	Higher
Computation	Efficient for feature extraction	Expensive for high-resolution images
Feature detection	Local patterns	Global pixel relationships
Translation handling	Better	Poorer
Scalability	Excellent	Limited
Training efficiency	High	Lower

3. Parameter Sharing

Suppose a CNN uses a **3×3 filter**.

The same filter can scan the entire image:

Image → Filter at Position 1 → Position 2 → Position 3 → ...

The filter does not need a new set of weights at every position.

For a grayscale image and 32 filters:

weights are required, excluding biases.

The same 288 weights are reused throughout the image.

4. Independent Weights

In a fully connected network, every input pixel is connected to every neuron with separate weights.

For an image of:

pixels and 1,000 neurons:

weights are needed for just one layer.

Thus, the memory and computational requirements become extremely high.

5. Feature Detection

Parameter sharing allows a CNN filter to detect the same feature wherever it appears.

For example, a **crack-detection filter** can identify a crack at:

- Top-left
- Center
- Bottom-right

without learning separate filters for each location.

6. Translation-Related Invariance

Pooling and convolution make CNNs more tolerant to small changes in object location.

For example:

Image 1: Crack appears on left side

Image 2: Crack appears on right side

The same filter can detect the crack in both locations.

This is particularly useful for surveillance images where objects constantly change position.

7. Critical Evaluation

Parameter sharing reduces:

- Number of parameters
- Memory consumption
- Training time
- Risk of overfitting

At the same time, it improves the ability to recognize repeated visual patterns across different locations.

Conclusion

For large-scale surveillance systems, CNN parameter sharing is significantly more efficient than independent weights.

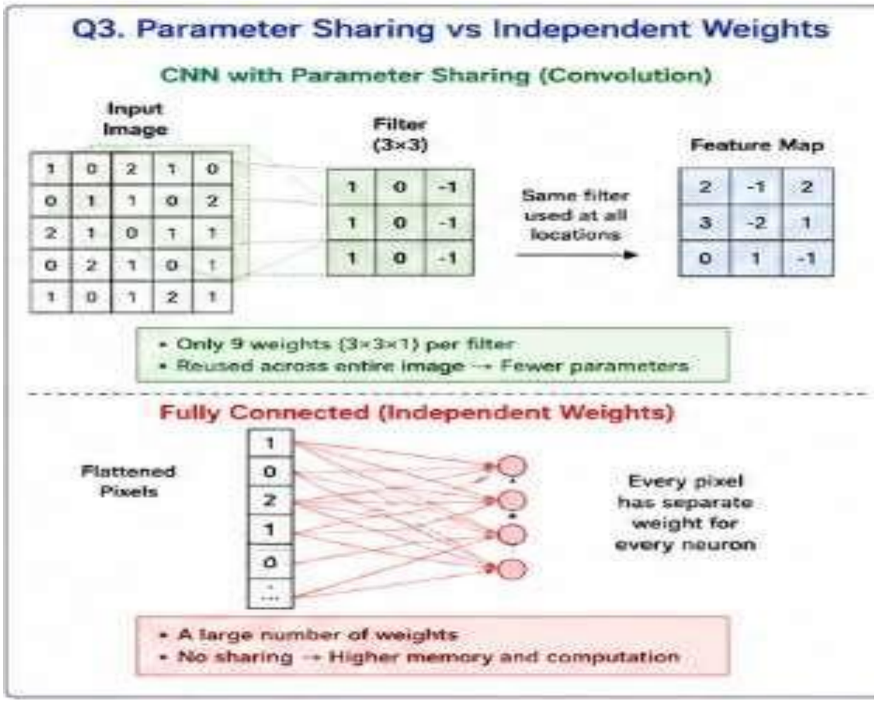


Figure 3: Parameter Sharing Vs Independent Weights

Therefore:

This makes CNNs highly suitable for processing thousands of high-resolution images.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering AlexNet and ResNet as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Solution:

1. Problem Understanding

AlexNet and ResNet are important CNN architectures used for image recognition. However, they differ significantly in depth, architecture, training strategy, parameter efficiency, and ability to handle very deep networks.

2. Comparison

Parameter	AlexNet	ResNet
Introduced	2012	2015
Depth	8 learnable layers	Often 18, 34, 50, 101, 152+ layers
Architecture	Conventional sequential CNN	Residual/skip connections
Main filters	Large early filters such as 11×11	Mostly 3×3 convolutions
Activation	ReLU	ReLU
Pooling	Max pooling	Pooling/strided convolution depending on stage
Parameters	About 60 million	Varies; ResNet-50 about 25.6 million
Training deep networks	Difficult	Easier
Vanishing gradient	More susceptible	Strongly reduced
Feature extraction	Good	Excellent
Computational cost	Moderate	Depends on depth
Complex recognition	Limited compared with modern deep models	Strong
Main innovation	Deep CNN + GPU training + ReLU	Residual learning

3. AlexNet Architecture

A simplified structure is:

Input → Conv → ReLU → Pool → Conv → ReLU → Pool → FC → FC → Output

AlexNet demonstrated that deep CNNs could achieve strong performance on large-scale image classification.

However, it uses a large number of parameters, particularly in its fully connected layers.

4. ResNet Architecture

ResNet introduced **residual/skip connections**.

Instead of learning:

the network learns:

and produces:

The shortcut connection allows information and gradients to flow more easily through very deep networks.

Simplified structure:

Input → Conv → Residual Block → Residual Block → Residual Block → Classification

5. Vanishing Gradient Problem

In very deep networks, gradients may become extremely small during backpropagation.

This can make early layers difficult to train.

ResNet addresses this using skip connections:

The identity path helps preserve information and gradients.

6. Feature Extraction

AlexNet learns:

ResNet can learn much deeper representations:

Thus, ResNet is generally better suited to complex recognition tasks.

7. Critical Evaluation

AlexNet advantages

- Simpler architecture
- Historically important
- Easier to understand
- Lower architectural complexity

AlexNet limitations

- Large parameter count
- Older architecture
- Less effective for highly complex modern tasks
- Limited depth

ResNet advantages

- Very deep architecture
- Skip connections
- Better gradient flow
- Strong feature extraction
- High classification performance

ResNet limitations

- Deeper versions can require substantial computational resources
- More complex architecture

8. Recommendation

For a **complex image recognition application**, **ResNet is recommended**.

The main reason is that residual connections make it possible to train much deeper networks without suffering as severely from the degradation and vanishing-gradient problems.

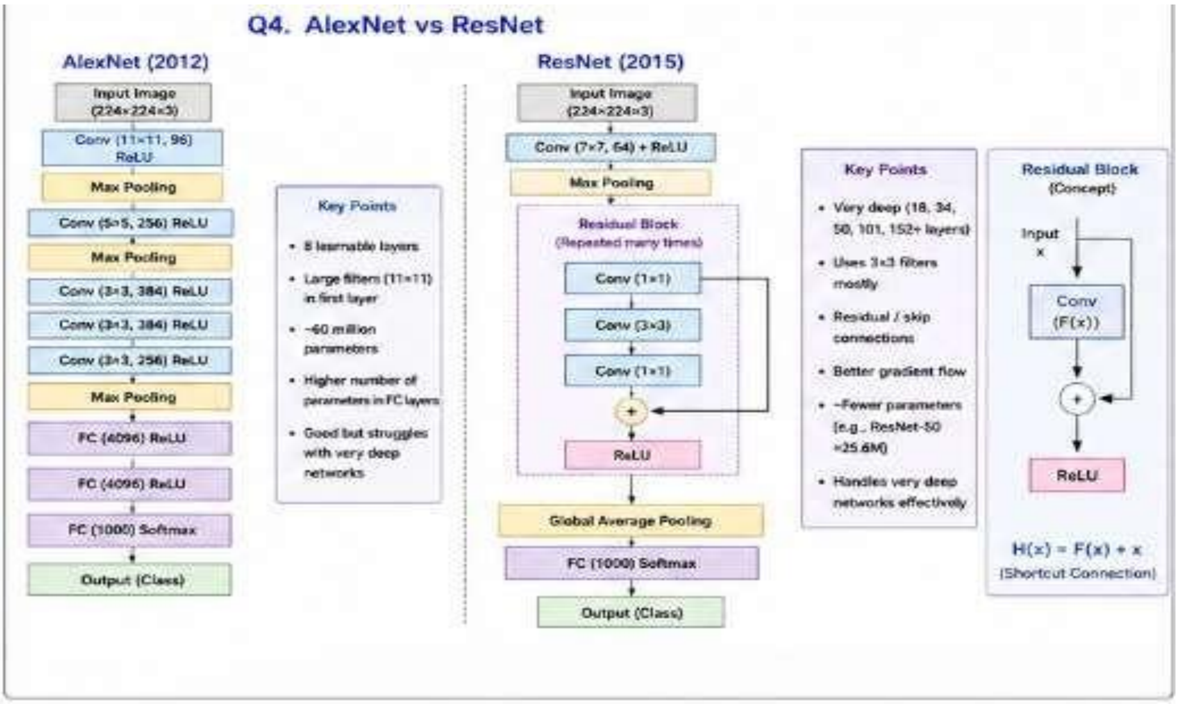


Figure 4: AlexNet Vs ResNet

Final Conclusion

ResNet provides deeper feature extraction, improved training stability, and generally better classification capability.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare **a** CNN without regularization and a CNN using dropout, data augmentation, and batch normalization in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

Solution:

1. Problem Understanding

The agricultural CNN achieves very high training accuracy but performs poorly on unseen leaf images.

This is a classic indication of **overfitting**.

Example:

Performance	**Training**	**Testing**
Accuracy	98%	70%

The model has learned the training images very well but cannot generalize to new images.

2. Comparison

Parameter	**Non-Regularized CNN**	**Regularized CNN**
Overfitting	High	Reduced
Generalization	Poor	Better
Training accuracy	Often very high	May be slightly lower
Testing accuracy	Often lower	Usually improved
Model robustness	Low	High
Data requirement	Higher	Data augmentation helps
Training stability	Can be unstable	Batch normalization improves stability
Complexity	Lower	Slightly higher

3. Dropout

Dropout randomly disables a proportion of neurons during training.

For example:

100 neurons → randomly deactivate 20 neurons → 80 active neurons

This prevents the network from becoming excessively dependent on particular neurons.

Benefit

For a leaf disease CNN, dropout can be applied before the final fully connected layers.

4. Data Augmentation

Data augmentation creates modified versions of existing training images.

Examples include:

- Rotation

- Horizontal flipping
- Scaling
- Cropping
- Translation
- Brightness variation

For example:

Original leaf → Rotation → Flip → Crop → Brightness change

This increases the effective diversity of the training data.

Importance

A leaf may appear:

- At different angles
- Under different lighting
- At different positions
- At different scales

Data augmentation helps the model learn disease characteristics rather than memorizing one particular image arrangement.

5. Batch Normalization

Batch normalization normalizes intermediate activations during training.

It helps:

- Stabilize training
- Improve gradient flow
- Allow faster convergence
- Reduce sensitivity to initialization

It can be represented conceptually as:

This makes CNN training more stable.

6. Critical Evaluation of Techniques

Dropout

Best for: Reducing dependence on specific neurons.

Advantage: Helps control overfitting.

Limitation: Can increase training time and may be unnecessary in every convolutional layer.

Data Augmentation

Best for: Increasing training diversity.

Advantage: Particularly valuable for agricultural images because real-world images vary in orientation and lighting.

Limitation: Unrealistic transformations can negatively affect learning.

For example, excessive image distortion may change the actual appearance of disease symptoms.

Batch Normalization

Best for: Training stability and faster convergence.

Advantage: Helps the network train efficiently.

Limitation: Adds additional operations and is not a direct replacement for all regularization techniques.

7. Recommended Approach

For plant disease classification, the recommended pipeline is:

Leaf Images

↓

Data Augmentation

↓

Convolution + ReLU

↓

Batch Normalization

↓

Pooling

↓

Deeper Convolution Layers

↓

Dropout

↓

Fully Connected Layer

↓

Disease Classification

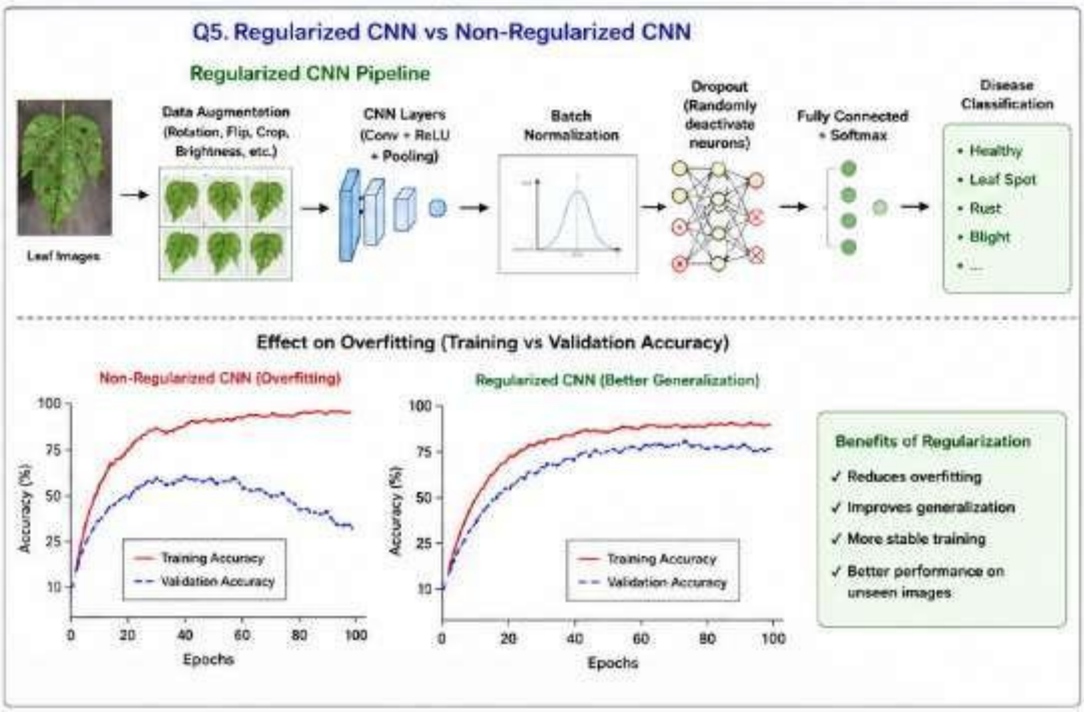


Figure 5: Regularized CNN Vs Non-Regularized CNN

8. Final Recommendation

A **regularized CNN** is clearly more suitable than a non-regularized CNN for this application.

The best approach is to combine the techniques rather than depend on only one:

This combination helps the model:

- Reduce overfitting
- Improve generalization
- Handle real-world image variations
- Train more reliably
- Improve performance on unseen leaf images

Overall Conclusion

For reliable real-world agricultural disease classification:

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generats meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand